

Ali Tleis

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Security Clearance: DoD Investigation in Progress

EDUCATION

Northeastern University

Boston, MA

Bachelor of Science in Computer Science

Expected May 2028

- **Relevant Coursework:** Algorithms & Data Structures, Object-Oriented Design, Artificial Intelligence, Discrete Structures, Programming in C++

EXPERIENCE

MIT Lincoln Laboratory

Lexington, MA

Web Application Developer (AI Integration) Co-op

Jul 2026 – Present

- Owning design and implementation of an AI-assisted search layer over the Laboratory's Apache Solr index, authoring the technical proposal and driving it through engineering review ahead of schedule
- Engineered retrieval logic that ranks and narrows candidate documents before model invocation, cutting query response time 70% and token consumption per request 40%
- Delivered a working end-to-end prototype and demoed it to the group, unifying content from SharePoint, internal documentation, and the public Laboratory site behind a single search interface
- Built the ingestion layer on Norconex, handling authenticated access, JavaScript-rendered pages, and mixed document formats to normalize content from multiple Laboratory web properties and SharePoint into the Solr index

Top Choice Realty

Staten Island, NY

Frontend Developer Intern

Jun 2025 – Sep 2025

- Architected reusable React/TypeScript component patterns to standardize dashboard and intake workflows, improving UI consistency and accelerating feature iteration
- Refactored MongoDB schema design to enforce data consistency across client and agent records, eliminating duplicate entries and improving query reliability
- Designed and implemented Python and C# automation pipelines to synchronize distributed MongoDB records across virtualized environments, reducing manual reconciliation time by 30%

Robert DeFalco Realty

Staten Island, NY

Computer Technician Intern

Jun 2023 – Sep 2023

- Developed PowerShell automation for workstation provisioning and software deployment across 20+ employee systems, standardizing Windows and Linux configurations and reducing setup time

PROJECTS

Eternal2x (DaVinci Resolve Smart Upscale) | *Python, Lua, OpenCV, FFmpeg*

Jan. 2026 – Present

- Engineered a DaVinci Resolve plugin that decodes footage through OpenCV's FFmpeg backend, computes per-frame motion scores via tile-based detail analysis, and groups high-motion frames into marked segments
- Automated cutting at marker positions and timeline regrouping, applying 2x upscale with Optical Flow interpolation on motion segments and Nearest on static segments for speed
- Shipped a one-click Windows and macOS installer that provisions dependencies and writes plugin configuration, with an auto-updater across tagged releases

Eternal Summary | *JavaScript, Chrome MV3, Node/Express, Docker, Fly.io, Gemini API*

Sep. 2023 – Present

- Built a Chrome extension (Manifest V3) using a background service worker and injected content scripts to extract live page text and return one-click AI-generated summaries
- Deployed a containerized Node/Express backend on Fly.io to proxy Gemini API calls, scoping host permissions to the backend origin and keeping credentials in runtime secrets

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Java, C#, C++

Frameworks: React, Next.js, Node.js/Express, Tailwind CSS, Vite, REST API Design

Data: MongoDB/Mongoose, PostgreSQL, MySQL, Apache Solr

Tools: Git, Docker, Linux, Jira, Confluence, Boomi, SharePoint, Fly.io, PowerShell, OpenCV, Gemini API